Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/08/2005
PI Name:	Balasubramaniam, Ramaswamy Ph.D.		
Project Title:	Instability of Miscible Interfaces		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Cleveland	**State:**	OH
Zip Code:	44135	**Congressional District:**	10
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	04/01/2000	**End Date:**	03/01/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Rashidnia, Nasser (National Center for Microgravity Research) Maxworthy , Tony (University of Southern California) Alexander , Iwan (Case Western Reserve University)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Experiments will be performed using density-matched fluids in order to simulate reduced gravity conditions. The fluids used will chiefly be silicone oils of various viscosities, with solvents added to them to achieve density matching. The viscosity ratio, the direction of the displacement with respect gravity, and the speed of the displacement process will be varied. The shape of the mixing front will be observed by a video camera or an interferometer. Direct numerical simulations will also be performed to obtain the conditions for which the interface is unstable. Time accurate three-dimensional simulations of the momentum and mass transport equations will be performed using the method of finite differences. We will include the effects of variable density and Korteweg stresses, which have been identified by Joseph and Renardy, 1993 (Fundamentals of Two-Fluid Dynamics, Springer-Verlag) to be important when the fluids are miscible. The objective of this project is to perform experimental and theoretical research on the dynamics of miscible fluids in a cylindrical tube, when a fluid is displaced by another fluid with a different viscosity. Preliminary experiments indicate that when the displacing fluid is more viscous, the mixing front can assume a sinuous shape. The specific goal is to identify conditions when such instabilities occur. It is widely believed that the dynamics of miscible fluids is important in porous media flows encountered in enhanced oil recovery, fixed bed regeneration, hydrology and filtration. The specific problem of the displacement of fluids where the more viscous fluid is surrounded by the less viscous one is encountered in water lubricated pipelining of crude oil.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Experiments on the displacement of a less viscous liquid by a more viscous liquid have been completed. The liquids used are silicone oils of various viscosities. The more viscous fluid has a higher density than the less viscous fluid. Both upward displacement and downward displacement has been studied. Our observations show that for downward displacement, the "interface' between the liquids can be unstable and exhibit an asymmetric, sinuous shape. During upward displacement, the interface forms an axisymmetric finger of the intruding liquid. At the tip of the main finger, a spike of the more viscous liquid is observed under some conditions. The Reynolds number is negligibly small in these experiments, and the Peclet number for mass transfer is O(50)-O(10^5). The influence of gravity is characterized by a dimensionless parameter F. A plot of F versus the viscosity ratio of the fluids has been constructed to delineate the stable and unstable regimes of flow. A manuscript describing the experimental results has been submitted for publication to Physics of Fluids.
Bibliography Type:	Description: (Last Updated: 03/01/2011)
Articles in Peer-reviewed Journals	Rashidnia N, Balasubramaniam R, Schroer RT. "The formation of spikes in the displacement of miscible fluids." Annals of the New York Academy of Sciences, 2004 Nov;1027:311-6. PMID:15644364 , Nov-2004
Articles in Peer-reviewed Journals	Balasubramaniam R, Rashidnia N, Maxworthy T, Kuang J. "Instability of miscible interfaces in a cylindrical tube." Physics of Fluids 2005 May;17(5):052103. http://dx.doi.org/10.1063/1.1884645 , May-2005